

Murillo F. Rodrigues

Computational Biologist

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Education

University of Oregon

*PhD in Biology**Eugene, OR, USA**Sept 2018 – Mar 2024*

- Dissertation: Understanding evolution with simulations: Three Tales about Trees
- Advisors: Dr. Andrew Kern & Dr. Peter Ralph

Universidade de São Paulo

*MSc in Genetics and Evolutionary Biology**São Paulo, SP, Brasil**Jan 2016 – Aug 2018*

- Thesis: Adaptive or neutral clines? Integrating genome-wide clinal and seasonal variation to infer natural selection in *Drosophila melanogaster*
- Advisor: Dr. Rodrigo Cogni

Universidade de São Paulo

*BS in Biological Sciences**São Paulo, SP, Brasil**Feb 2012 – Dec 2015*

- Thesis: Diversity and phylogenetic positioning of freshwater lineages of *Rhinebothrhum* from Lake Maracaibo and Orinoco basin, Venezuela

Academic Positions

Postdoctoral Researcher

*Oregon Health & Science University**Portland, OR, USA**May 2024 – present*

- Supervisors: Dr. Jeff Wall & Dr. Don Conrad

Publications

- Inbreeding and deleterious variation in South Asian populations** 2025
 Jeffrey D Wall, *Murillo F Rodrigues*
[10.1038/s41598-025-18348-5](https://doi.org/10.1038/s41598-025-18348-5) (Scientific Reports)
- Gene flow weakens genomic clines while selection maintains adaptive loci: two decades of evolution in *Drosophila melanogaster* natural populations** 2025
 † Equal contribution
 Vitória Horvath Miranda, Tiago Ribeiro, Joel M Alves, Walter Eanes, Evgeny Brud, Krishna Veeramah, John Pool, *Murillo F Rodrigues*†, Rodrigo Cogni†
[10.1101/2025.11.28.691231](https://doi.org/10.1101/2025.11.28.691231) (bioRxiv)
- Bridging forward-in-time and coalescent simulations using pyslim** 2025
 † Equal contribution
 Shyamalika Gopalan†, *Murillo F Rodrigues*†, Peter L Ralph, Benjamin C Haller
[10.1101/2025.09.30.679676](https://doi.org/10.1101/2025.09.30.679676) (bioRxiv)
- SimHumanity: Using SLiM 5.0 to run whole-genome simulations of human evolution** 2025
 Benjamin C Haller, Chase W Nelson, *Murillo F Rodrigues*, Philipp W Messer
[10.1101/2025.09.01.673541](https://doi.org/10.1101/2025.09.01.673541) (bioRxiv)
- Accessible, realistic genome simulation with selection using stdpopsim** 2025
 † Equal contribution
 Graham Gower†, Nathaniel S Pope†, *Murillo F Rodrigues*†, Silas Tittes†, Linh N Tran†, Ornob Alam, Maria Izabel A Cavassim, Peter D Fields, Benjamin C Haller, Xin Huang, et al.
[10.1093/molbev/msaf236](https://doi.org/10.1093/molbev/msaf236) (Molecular Biology and Evolution)
- Neofunctionalization driven by positive selection led to the retention of the *logs2* gene encoding an *Aedes* specific dsRNA binding protein** 2024
 Carlos F Estevez-Castro, *Murillo F Rodrigues*, Antinéa Babarit, Flávia V Ferreira, Elisa G de Andrade, Eric Marois, Rodrigo Cogni, Eric RGR Aguiar, João T Marques, Roenick P Olmo
[10.1186/s12915-024-01821-4](https://doi.org/10.1186/s12915-024-01821-4) (BMC Biology)
- Shared evolutionary processes shape landscapes of genomic variation in the great apes** 2024
Murillo F Rodrigues, Andrew D Kern, Peter L Ralph
[10.1093/genetics/iyae006](https://doi.org/10.1093/genetics/iyae006) (Genetics)

- Expanding the stdpopsim species catalog, and lessons learned for realistic genome simulations** 2023
M Elise Lauterbur, Maria Izabel A Cavassim, Ariella L Gladstein, Graham Gower, Nathaniel S Pope, Georgia Tsambos, Jeffrey Adrion, Saurabh Belsare, Arjun Biddanda, Victoria Caudill, et al.
[10.7554/elife.84874](https://doi.org/10.7554/elife.84874) (eLife)
- Efficient ancestry and mutation simulation with msprime 1.0** 2022
Franz Baumdicker, Gertjan Bisschop, Daniel Goldstein, Graham Gower, Aaron P Ragsdale, Georgia Tsambos, Sha Zhu, Bjarki Eldon, E Castedo Ellerman, Jared G Galloway, et al.
[10.1093/genetics/iyab229](https://doi.org/10.1093/genetics/iyab229) (Genetics)
- Clinal and seasonal changes are correlated in *Drosophila melanogaster* natural populations** 2021
Murillo F Rodrigues, Maria D Vibranovski, Rodrigo Cogni
[10.1111/evo.14300](https://doi.org/10.1111/evo.14300) (Evolution)
- Genomic responses to climate change: making the most of the *Drosophila* model** 2021
Murillo F Rodrigues, Rodrigo Cogni
[10.3389/fgene.2021.676218](https://doi.org/10.3389/fgene.2021.676218) (Frontiers in Genetics)
- Widespread selection and gene flow shape the genomic landscape during a radiation of monkeyflowers** 2019
Sean Stankowski, Madeline A Chase, Allison M Fuiten, Murillo F Rodrigues, Peter L Ralph, Matthew A Streisfeld
[10.1371/journal.pbio.3000391](https://doi.org/10.1371/journal.pbio.3000391) (PLoS Biology)

Fellowships & Grants

- Genetics Training Grant, CAS, University of Oregon (2019 – 2021)
- Research Internship Abroad Fellowship, The São Paulo Research Foundation (2017 – 2018)
- Master's Fellowship, The São Paulo Research Foundation (2016 – 2018)
- Undergraduate Research Fellowship, The São Paulo Research Foundation (2013 – 2014)

Presentations

- American Society of Primatologists Annual Meeting** *IL, USA*
Organized by the American Society of Primatologists *Sept 2025*
- Talk: A complete genomic resource for captive marmosets: insights into recent population history and biology
- Biology of Genomes** *NY, USA*
Organized by Cold Spring Harbor Laboratory *May 2025*
- Poster: A complete genomic resource for captive marmosets: insights into recent population history and biology
- SMBE2024** *Puerto Vallarta, Mexico*
Organized by the Society for Molecular Biology and Evolution *July 2024*
- Talk: A deep learning framework for evolutionary inference using ancestral recombination graphs
- SMBE2023** *Italy*
Organized by the Society for Molecular Biology and Evolution *July 2023*
- Talk: Shared evolutionary processes shape genomic variation in the great apes. [See slides here.](#)
- Population, Evolutionary, and Quantitative Genetics Conference** *CA, USA*
Organized by the Genetics Society of America *June 2022*
- Poster: Why are landscapes of diversity correlated in the great apes? [See poster here.](#)
- SMBEv2021** *Virtual*
Organized by the Society for Molecular Biology and Evolution *July 2021*
- Poster: Natural selection and landscapes of diversity in the great apes.
- Probabilistic Modeling in Genomics** *Virtual*
Organized by Cold Spring Harbor Laboratories *Apr 2021*
- Poster: Natural selection and landscapes of diversity in the great apes. [See poster here.](#)
- The Allied Genetics Conference** *Virtual*
Organized by the Genetics Society of America *Apr 2020*
- 8th Workshop on Cestode Systematics and Phylogeny** *SP, Brasil*
Organized by Department of Zoology, Universidade de São Paulo *July 2014*

Teaching

- Guest Lecturer** *Mar 2025*
Lewis & Clark College
Phylogenetics and Molecular Evolution – Guest lecture on inferring selection from phylogenies (2h)

Teaching Assistant <i>University of Oregon</i> Introduction to Programming for Biologists – Teaching assistant for consecutive terms (12h/week for 10 weeks)	<i>Jan 2019 – Mar 2020</i>
Teaching Assistant <i>University of Oregon</i> General Biology III: Populations – Teaching assistant (12h/week for 10 weeks)	<i>Sept 2018 – Dec 2018</i>
Teaching Assistant <i>Universidade de São Paulo</i> Molecular Ecology – Teaching assistant (6h/week for 16 weeks)	<i>Feb 2017 – June 2017</i>
Teaching Assistant <i>Universidade de São Paulo</i> Evolutionary Processes – Teaching assistant (6h/week for 16 weeks)	<i>Feb 2016 – June 2016</i>
Instructor <i>Universidade de São Paulo</i> Introduction to Biostatistics – Short course in Semana Temática da Biologia, IB/USP (12h)	<i>Oct 2015</i>
Undergraduate Teaching Assistant <i>Universidade de São Paulo</i> Introduction to Statistics – Undergraduate teaching assistant (6h/week for 16 weeks)	<i>Aug 2015 – Dec 2015</i>
Undergraduate Teaching Assistant <i>Universidade de São Paulo</i> Introduction to Systematics and Biogeography – Undergraduate teaching assistant (6h/week for 16 weeks)	<i>Feb 2013 – June 2013</i>

Mentoring

- Vitória Miranda, MSc, Genetics and Evolutionary Biology, Universidade de São Paulo (2022 – 2025) – Mentor and thesis committee member
- Carlos Henrique Passos, PhD, Genetics and Evolutionary Biology, Universidade de São Paulo (2025 – present) – Dissertation committee
- Rafael Viana, PhD, Ecology, Universidade de São Paulo (2025) – Defense committee

Software

tskit

Tree sequence toolkit for population genetics – *Contributor*

msprime

Coalescent simulation of ancestral histories and DNA sequences – *Contributor*

stdpopsim

Standard population genetic simulations library – *Core developer*

pyslim

Python interface for SLiM tree sequences – *Contributor*

scikit-allel

Python library for exploratory analysis of genetic variation data – *Contributor*

Service

- Peer Reviewer: GENETICS, Molecular Biology and Evolution, Molecular Ecology Resources, G3, Proceedings of the Royal Society B: Biological Sciences
- Treasurer, Graduate Evolutionary Biology and Ecology Students – A student-led organization providing career-building activities and community outreach

Scholarships & Awards

- Harvey E Lee Graduate Scholarship, University of Oregon (2022 – 2023)
- Marthe E. Smith Memorial Science Scholarship, CAS, University of Oregon (2022 – 2023)
- Hill Fund Award, CAS, University of Oregon (2019 – 2020)

Skills

Coding: Python, R, Bash, C++

Computing: Unix, HPC, SLURM, Snakemake (workflow management), Git

Bioinformatics: Next-gen sequencing data quality control and preprocessing, genome and transcriptome alignment, SNP calling, differential expression analysis

Statistics: Frequentist and Bayesian statistics (using R and Stan), simulation-based inference, machine learning (mostly using PyTorch – CNN, RNN and GNN)